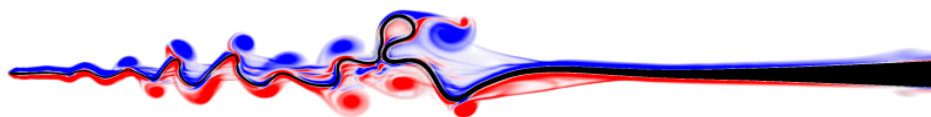


Sílvio Cândido

Curriculum Vitae



Mechanical Engineer

Organisation européenne pour la recherche nucléaire, CERN
Geneva, Switzerland

✉ silvio.candido@cern.ch
🌐 Webpage @ github
🆔 Orcid

EDUCATION

Ph.D. in Mechanical Engineering 2019-2024

University of Beira Interior, Portugal

[Thesis: Atomization of Electrohydrodynamic Jets - modulation, performance and applications.](#)

M.Sc. in Electromechanical Engineering 2017-2019

University of Beira Interior, Portugal

[Dissertation: Numerical studies about multiphase uniformity of the flow inside mixing chambers using CFD – Analysis of the chamber of the CLOUD experience at CERN.](#)

B.Sc. in Electromechanical Engineering 2014-2017

University of Beira Interior, Portugal

EXPERIENCE

Mechanical Engineer Fellow 04/2024-today

European Organization for Nuclear Research (CERN), Switzerland

Developing beam interception devices for next-generation particle accelerators, focusing on liquid lead systems for energy absorption and particle generation. Employed computational fluid dynamics to model magnetohydrodynamic effects, thermomechanical stresses through conjugate heat transfer simulations, and explicit dynamics.

CFD Researcher & ML Specialist 01/2024-03/2024

Center for Mechanical and Aerospace Science and Technologies (C-MAST), Portugal

Conducting research on Electrospray formation through multiphysics simulations and developing Machine Learning models for Digital Twin applications.

Doctoral Project Researcher 01/2021-12/2023

Fundação para a Ciência e a Tecnologia (FCT), Portugal

Doctoral research on electrohydrodynamic effects in liquid jets for electrospray formation. Achieved over 95% accuracy in free-surface simulations.

[Grant no 2020.04517.BD]

Project Researcher 09/2019-12/2020

University of Beira Interior (UBI), Portugal

Research in complex multi-physics flow, more specifically in electro-hydrodynamic flows for electrospray formation.

[Project IndTech 4.0 – POCI-01-0247-FEDER-026653]

Internship Experience

- Doctoral Research Internship** 05/2023
under ERASMUS+, Universitatea de Vest din Timisoara Facultatea de Fizica, Romania
Short-term doctoral mobility scholarship developing work in Lattice Boltzmann Methods for Fluid dynamics and also GPU computations for parallel acceleration of complex calculations.
- Researcher trainee** 08/2017
University of Beira Interior, Portugal
Computational modulation of a disc pump, type Tesla. Utilization of the tools of Solidworks Flow Simulation for calculating the characteristics of the 3D flow in the disc pump.

SCIENTIFIC ACTIVITIES

Journal Publications

- PP1 S. Cândido, R. Franqueira Ximenes, A Lechner *et al.* Design and modeling of a liquid-lead dump concept for beamstrahlung radiation absorption in the CERN Future Circular e^+e^- Collider, (2026). [\[DOI\]](#) (arXiv preprint, under review at *Physical Review Accelerators and Beams*.)
- J7 Boscolo, M., Palla, F., Ammirabile, G. *et al.* Status of the FCC-ee interaction region design, *EPJ Techniques and Instrumentation* 12, 4 (2025). [\[DOI\]](#)
- J6 M. Pendar, S. Cândido, J. Páscoa, R. Lima, *Enhancing Automotive Paint Curing Process Efficiency: Integration of Cfd and Variational Auto-Encoder Techniques* **SSRN 4715982**, (2025)
- J5 S. Cândido, J. Páscoa, *Data-driven surrogate modelling of multistage Taylor cone-jet dynamics* **Physics of Fluids**, (2024) [\[DOI\]](#)
- J4 M. Pendar, S. Cândido, J. Páscoa, *Optimization of Painting Efficiency Applying Unique Techniques of High-voltage Conductors and Nitrotherm Spray: Developing Deep Learning Models Using Computational Fluid Dynamics Dataset*, **Physics of Fluids** (2023) [\[DOI\]](#)
- J3 S. Cândido, J. Páscoa, *On modal decomposition as surrogate for charge-conservative EHD modelling of Taylor Cone jets*, **International Journal of Engineering Science**, (2023) [\[DOI\]](#)
- J2 S. Cândido, J. Páscoa, *Dynamics of three-dimensional electrohydrodynamic instabilities on Taylor cone jets using a numerical approach*, **Physics of Fluids** (2023) [\[DOI\]](#)
- J1 M. Moreira, F. Rodrigues, S. Cândido, J. Páscoa, G. Santos, *Development of a Background Oriented Schlieren (BOS) system for thermal characterization of plasma actuators induced flow*, **Energies** (2022) [\[DOI\]](#)

Peer-Reviewed Conference Proceedings

- CP17 P. Jurj, S. Candido, et al., Refined design of the front-end complex for a Muon Cooling Demonstrator at CERN, *IPAC'26 - International Particle Accelerator Conference*, Deauville, France. 2026. [\[DOI\]](#)
- CP16 S. Candido, et al., Conceptual comparison of liquid lead flow configurations for a muon collider, *IPAC'26 - International Particle Accelerator Conference*, Deauville, France. 2026. [\[DOI\]](#)

- CP15 J. Manczak, S. Candido, et al., FLUKA studies of spent beam extraction in the muon collider target system, *IPAC'26 - International Particle Accelerator Conference*, Deauville, France. 2026. [\[DOI\]](#)
- CP14 S. Candido, et al., Comparative conceptual designs of liquid lead absorbers for the FCC-ee beamstrahlung dump, *IPAC'26 - International Particle Accelerator Conference*, Deauville, France. 2026. [\[DOI\]](#)
- CP13 S. Candido, et al., Magnetohydrodynamic effects in liquid lead target concept for Muon Colliders, *IPAC'25 - the 16th International Particle Accelerator Conference*, Taiwan. 1-6 June, 2025. [\[DOI\]](#)
- CP12 J. Manczak, S. Candido, et al., New developments in the design of the muon production target area of a multi-TeV muon collider, *IPAC'25 - the 16th International Particle Accelerator Conference*, Taiwan. 1-6 June, 2025. [\[DOI\]](#)
- CP11 S. Candido, et al., Conceptual design and optimization of a liquid lead circuit as beamstrahlung absorber for the CERN's FCC, *IPAC'25 - the 16th International Particle Accelerator Conference*, Taiwan. 1-6 June, 2025. [\[DOI\]](#)
- CP10 S. Candido, M. Pendar, J. Páscoa, *Improving Efficiency of Automotive Coating and Curing Processes Through Deep Learning Algorithms and High-Fidelity CFD Modeling*, International Mechanical Engineering Congress and Exposition, New Orleans, LA. October 29 — November 2, 2023. [\[DOI\]](#)
- CP9 S. Candido, J. Páscoa, *A Three-Dimensional Numerical Investigation of Taylor Cone Jet Instabilities Using the VOF Method* International Mechanical Engineering Congress and Exposition, New Orleans, LA. October 29 — November 2, 2023. [\[DOI\]](#)
- CP8 S. Candido, J. Páscoa, *Numerical Simulation of Axisymmetric Electrohydrodynamic Jets with Volume of Fluid Method*, ICEUBI - International Congress on Engineering, Covilhã, Portugal, November 28, 29 and 30, 2022. Paper no. 5527
- CP7 M. Moreira, F. Rodrigues, J. Páscoa, S. Candido, *Development and validation of a background oriented Schlieren (BOS) system for air density and temperature quantification*, ICEUBI - International Congress on Engineering, Covilhã, Portugal, November 28, 29 and 30, 2022.
- CP6 S. Cândido, J. Páscoa, *Numerical Analysis of Interfacial Electrohydrodynamic Flow With Modal Decomposition*, International Mechanical Engineering Congress and Exposition, Columbus, Ohio. October 30 — November 3, 2022. [\[DOI\]](#)
- CP5 S. Cândido, J. Páscoa, *Numerical Simulation of Electrified Liquid Jets Using a Geometrical VoF Method*, International Mechanical Engineering Congress and Exposition, Virtual, Online. 2021. [\[DOI\]](#)
- CP4 S. Cândido, J. Páscoa, *Numerical Analysis on the Stability Conditions of an Electrohydrodynamic Jet*, International Mechanical Engineering Congress and Exposition, Portland, Oregon. 2020. [\[DOI\]](#)
- CP3 S. Cândido, J. Páscoa, A. Tomé, A. Amorim, and S. Weber, *CFD Analysis of Flowstructures in a Mixing Chamber*, International Mechanical Engineering Congress and Exposition, Salt Lake City. November 11-14, 2019. [\[DOI\]](#)
- CP2 S. Cândido, J. Páscoa, *3D unsteady RANS computation of the mixing on a T-junction*,

- International Congress on Engineering, Covilhã, November 27-29, 2019. [\[DOI\]](#)
- CP1 J. C. Pascoa, [S. Candido](#), F. Charrua-Santos, A. Espirito-Santo and M. Canario, *Disc Turbine for Energy Harvesting*, International Mechanical Engineering Congress and Exposition, Pittsburgh, November 9-15, 2018. [\[DOI\]](#)

Extended Abstract Conference Proceedings

- EA1 [S. Candido](#) and J. C. Pascoa, *Multi-physical Modelling of Electrohydrodynamic Liquid Jets: Perspectives on Deep Learning Surrogates*, The 9th European Congress on Computational Methods in Applied Sciences and Engineering, Lisbon, 3-7 June, 2024.

Posters

- P2 [S. Cândido](#), J. Páscoa, *Numerical Investigations on Atomization of Electrohydrodynamic Jets*, Encontro UBI Ciência e Inovação 2023, Covilhã, Portugal. [\[PDF\]](#)
- P1 [S. Cândido](#), J. Páscoa, *Advances on Modelling the Atomization of Electrohydrodynamic Jets*, Encontro Ciência 2023 - 5-7 July 2023, Aveiro, Portugal. [\[PDF\]](#)

Reviewer Activity

Reviewer for international journals including: Physics of Fluids (19); Journal of Aerosol Science (5); Journal of Fluid Mechanics (2); Fluids (2); Computation (1); Communications Engineering Nature (1).

Conference Activities

- **Conference Session Co-Chair**, at IMECE2023 , ASME (USA)

Session co-chair at the International Mechanical Engineering Congress and Exposition for the topic: Aerodynamics & Novel Aerospace Propulsion Systems

- **Conference Session Co-Chair**, at IMECE2021 , ASME (USA)

Session co-chair at the International Mechanical Engineering Congress and Exposition for the topic: Aerodynamics & Novel Aerospace Propulsion Systems

PROJECTS

(PI) Advanced Computing Project for Research and Innovation - A2: Investigations on Electrohydrodynamic Atomization: High-Fidelity Simulations of Taylor Cone Jet Atomization for Optimized Electrospray Processes. [\[DOI\]](#)

Access to 320 000 CPU core.horas of a High Performance Computing (HPC) from the National Network for Advanced Computing (RNCA) of Portugal.

Founded by: Fundação para a Ciência e a Tecnologia (FCT), 06/2024 to 05/2025 (12 Months). Value of 3259€

(PI) Advanced Computing Project for Research and Innovation - A0: High-Fidelity Simulation of Atomization of Taylor Cone Jets for Electrospinning and Electrospray. [\[DOI\]](#)

Access to 40 000 CPU core.hour of a High Performance Computing (HPC) from the National Network for Advanced Computing (RNCA) of Portugal.

Founded by: Fundação para a Ciência e a Tecnologia (FCT), 07/2023 to 12/2023 (6 Months)

COMMUNICATIONS

Invited Oral Presentation

- IP5 *Lead Target Jet and Curtain in Muon Production and Target Technologies* parallel session, 5th Annual Meeting of the International Muon Collider Collaboration & 3rd MuCol Annual Meeting
CERN, Geneva, Switzerland, June 2026
- IP4 *Graphite Target in Muon Production and Target Technologies* parallel session, 5th Annual Meeting of the International Muon Collider Collaboration & 3rd MuCol Annual Meeting
CERN, Geneva, Switzerland, June 2026
- IP3 *Development of Liquid Pb Technologies for Beam Intercepting Devices at CERN: A Case Study for Muon Collider Target and FCCee Beamstrahlung Absorber*, in Global Symposium on Lead and Lead Alloy Cooled Nuclear Energy Science and Technology. [\[Link\]](#)
30 September 2024. ENEA – Brasimone Research Centre, Italy.
- IP2 *Numerical Modelling of the Atomization of Electrohydrodynamic Jets*, in Seminars of the Research Center in Theoretical Physics. [\[Link\]](#)
11 May 2023. West University of Timisoara (Timisoara, România).
- IP1 *Talk on Environmental Sustainability*
In collaboration with the Social Action Services of the University of Beira Interior.
10th March 2020. University of Beira Interior, Covilhã, Portugal.

Oral Conference Presentation

- OP7 *Improving Efficiency of Automotive Coating and Curing Processes Through Deep Learning Algorithms and High-Fidelity CFD Modeling*
International Mechanical Engineering Congress and Exposition, New Orleans, LA.
October 30 — November 3, 2023. Columbus, Ohio, USA.
- OP6 *Numerical Analysis of Interfacial Electrohydrodynamic Flow With Modal Decomposition*
in IMECE - International Mechanical Engineering Congress and Exposition
October 30 — November 3, 2023. Columbus, Ohio, USA.
- OP5 *Numerical Simulation of Axisymmetric Electrohydrodynamic Jets with Volume of Fluid Method*, in ICEUBI - International Congress on Engineering
29th November, 2022. Covilhã, Portugal
- OP4 *Numerical Analysis of Interfacial Electrohydrodynamic Flow With Modal Decomposition*
in IMECE - International Mechanical Engineering Congress and Exposition
October 30 — November 3, 2022. Columbus, Ohio, USA.
- OP3 Numerical Simulation of Electrified Liquid Jets Using a Geometrical VoF Method,
in IMECE - International Mechanical Engineering Congress and Exposition
2021. Virtual, Online.
- OP2 Numerical Analysis on the Stability Conditions of an Electrohydrodynamic Jet,
in IMECE - International Mechanical Engineering Congress and Exposition, Online
2020. Virtual, Online.

OP1 3D unsteady RANS computation of the mixing on a T-junction,
in *International Congress on Engineering, Covilhã*
November 27-29, 2019. Covilhã, Portugal.

GRANTS, AWARDS

2021/23 **Ph.D. Grant** at Center for Mechanical and Aerospace Science and Technologies
(C-MAST), Covilhã, Portugal
Individual grant for 3 year doctoral studies from the the Portuguese Foundation for
Science and Technology (FCT) FCT grant no. 2020.04517.BD

2020 **Merit Scholarship**, DGES, Portugal
(Top of the class) Scholarship due to the classifications of the curricular units obtained in
the academic year 2018/2019.

CERTIFICATIONS, SKILLS AND COMPLEMENTARY FORMATION

Certifications

04/2023 **Lean Six Sigma Yellow Belt**
The Lean Six Sigma Company Portugal.

07/2018 **Certified SolidWorks Associate - CSWA**
Certification from the Dassault Systèmes for Mechanical Design in SolidWorks.

Complementary Formation

05/2024 **Structural mechanics simulation with Ansys Mechanical**
CADFEM France & Suisse 40h Hands-on Course.

11/2022 **Power BI**
Instituto CRIAP, Portugal 31h e-learning Course.

02/2022 **Hands on Machine Learning for Fluid Dynamics**
von Karman Institute for Fluid Dynamics, Belgium 2 ECTS e-learning course.

02/2022 **Machine Learning, Maths & Ethics: Hands-on (MOOC)**
Instituto Superior Técnico, Portugal E-learning course.

01/2022 **Energy sustainability**
DECO - Portuguese Association for Consumer Protection, Portugal E-learning Course.

11/2021 **Introduction to SQL**
DataCamp 4h E-learning Course.

05/2018 **Python Fundamentals & Data Science**
CFIUTE, University of Beira Interior, Portugal Professional course of 28:00 hours.

12/2017 **Python Fundamentals for engineering applications**
CFIUTE, University of Beira Interior, Portugal Professional course of 20:00 hours.

Languages

Portuguese: Native English: C1 French: A2

PROFESSIONAL MEMBERSHIP

2022/24 **Student Fellow** at ASME: The American Society of Mechanical Engineers
..... Member Number: 000103654845

VOLUNTEER EXPERIENCE

Long-term activities

- 2022/23 **Vice-President Fiscal Council**, Happy Wish Junior Initiative (HW), Portugal
Supervising the financial management of the HW Junior Initiative and performing a report about financial activities.
- 2021/24 **Finance Department Associate**, Happy Wish Junior Initiative (HW), Portugal
Support the financial management of the HW Junior Initiative.
- 2019/24 **Volunteer** of ReFood Covilhã, *Covilhã (Portugal)*
Volunteer activities during 2h every week. Being in charge of the facilities, receiving food donations from restaurants, and taking care of the appropriate food storage.
- 2016/17 **Treasurer** of Rotaract Club de Tavira, Tavira (Portugal)
Responsible for maintaining club financial records and dues payments for administrative service accounts.
- 2013/14 **Founding fellow** of Rotaract Club de Tavira, Tavira (Portugal)
Organizing and participating in fundraising events for support of local associations and international movements (e.g. End Polio Now).
- 2012/14 **Fellow Volunteer** of Interact Club de Tavira, Tavira (Portugal)
Organizing and participating in fundraising events to support local and international associations (e.g. End Polio Now). Participation in national conferences and leadership training events.

Short-term activities

- 03/2023 **Tutor at STEAM junior academy**, at AJSTEAM UBI, Covilhã (Portugal)
Guide and support young people (high-school) in the closing activities of the UBI STEAM academy. For two days (30h), with organization and orientation activities.

Updated June 2026